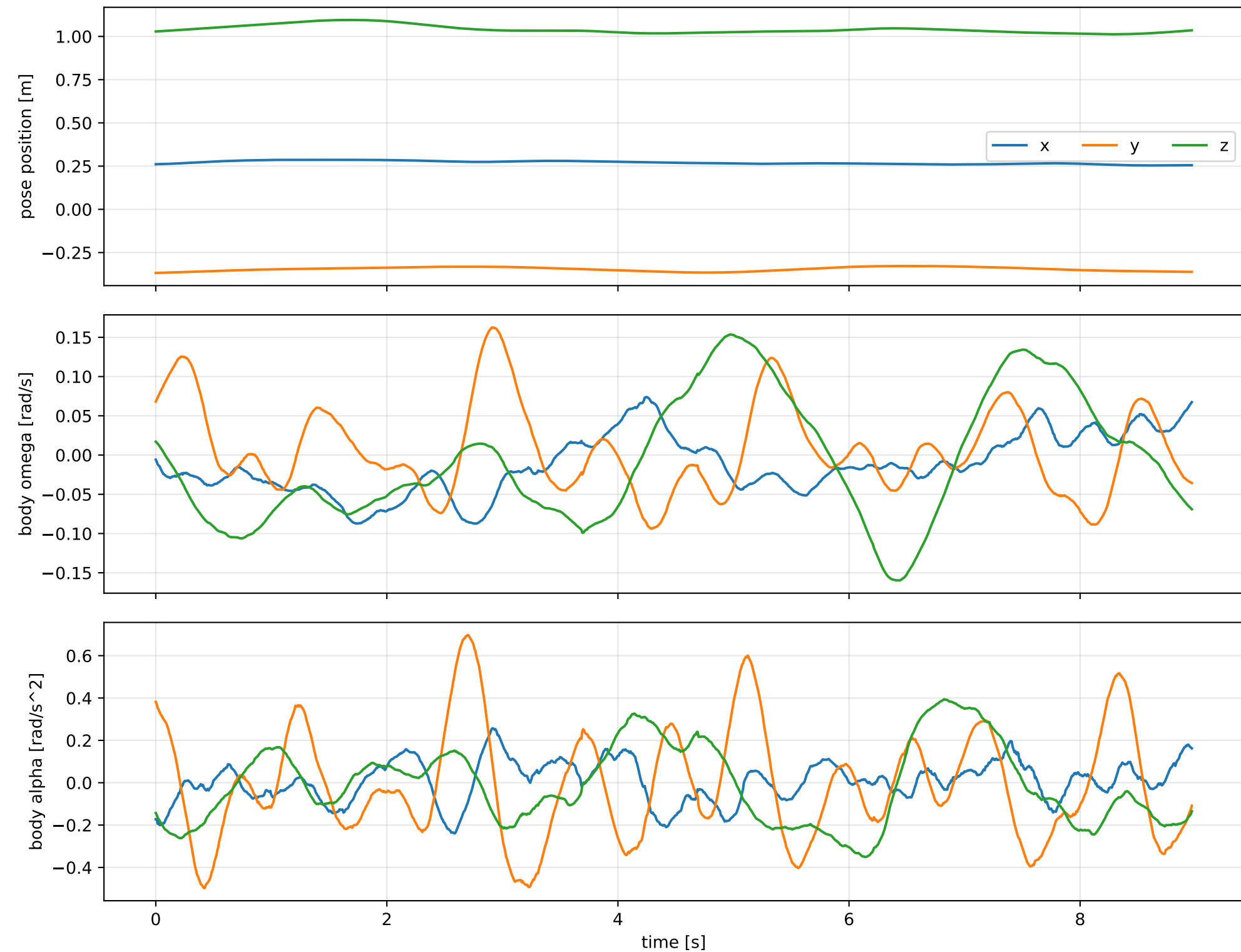
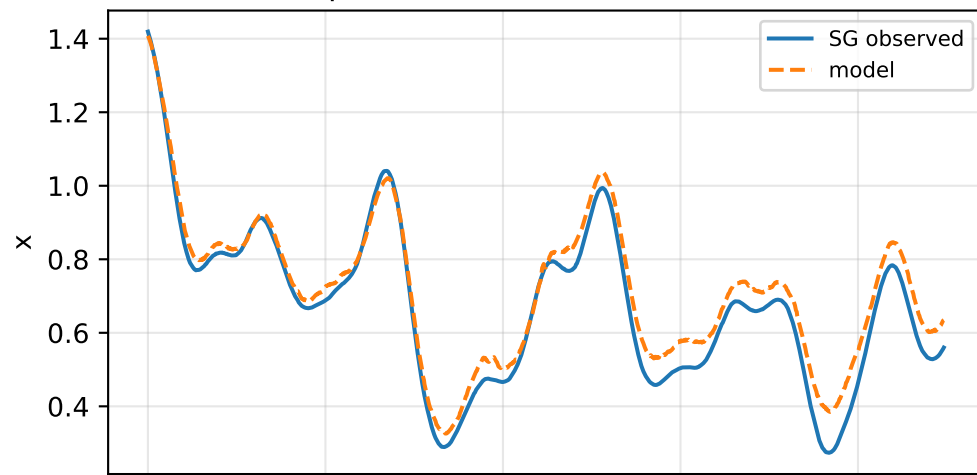


lag_legacy_smooth_schedule: SG trajectory and derived motion

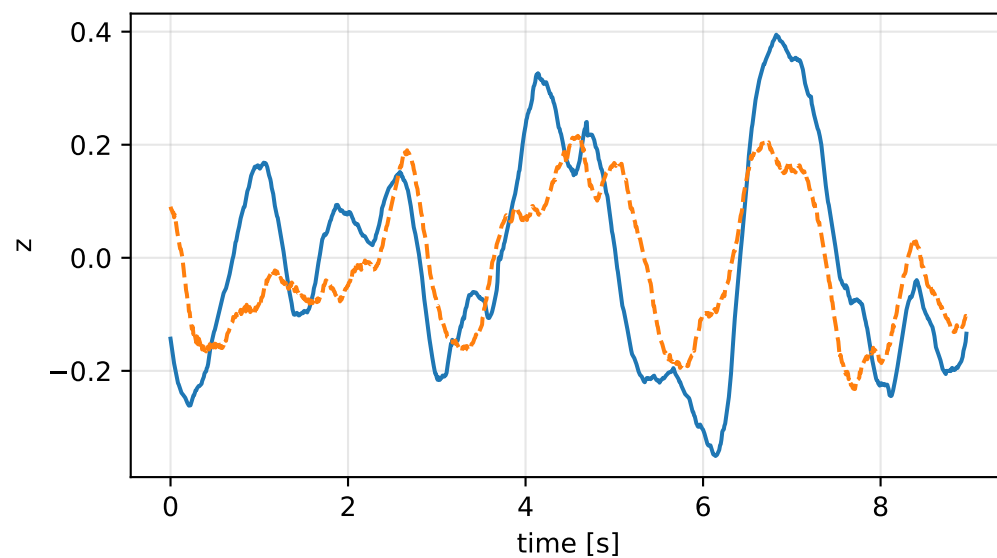
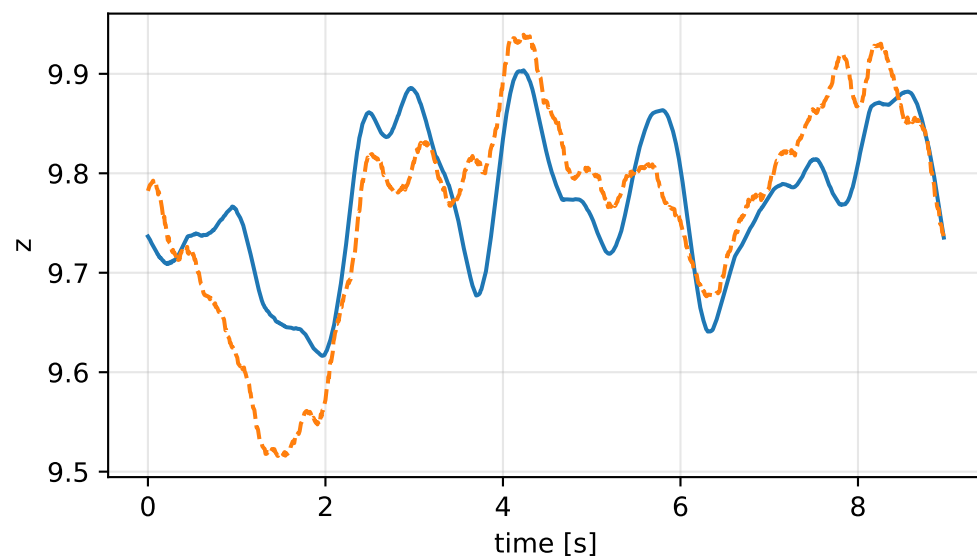
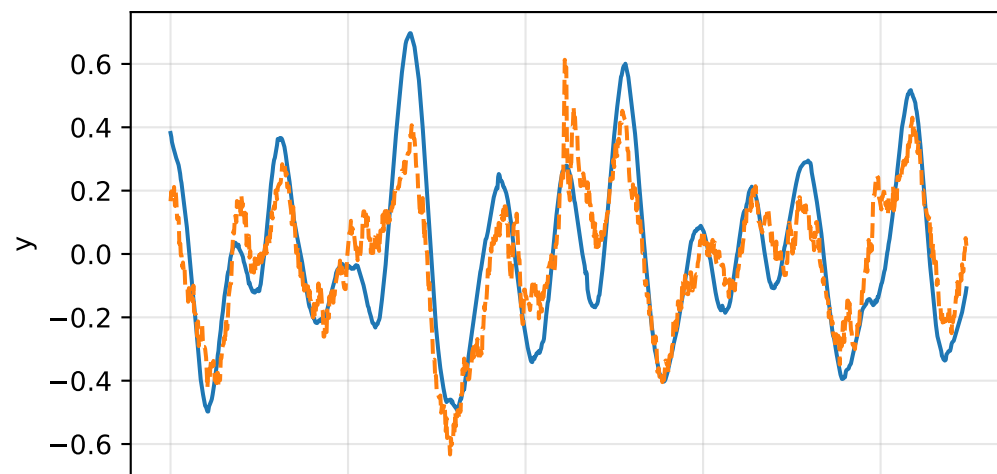
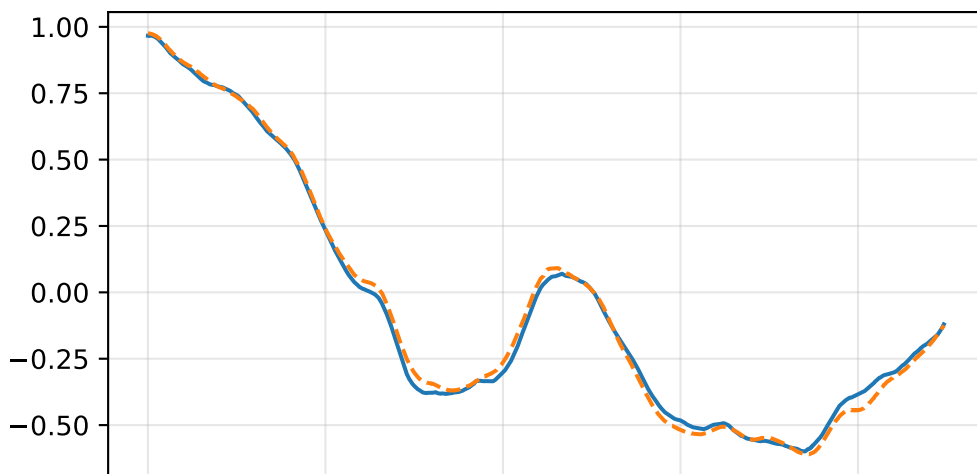
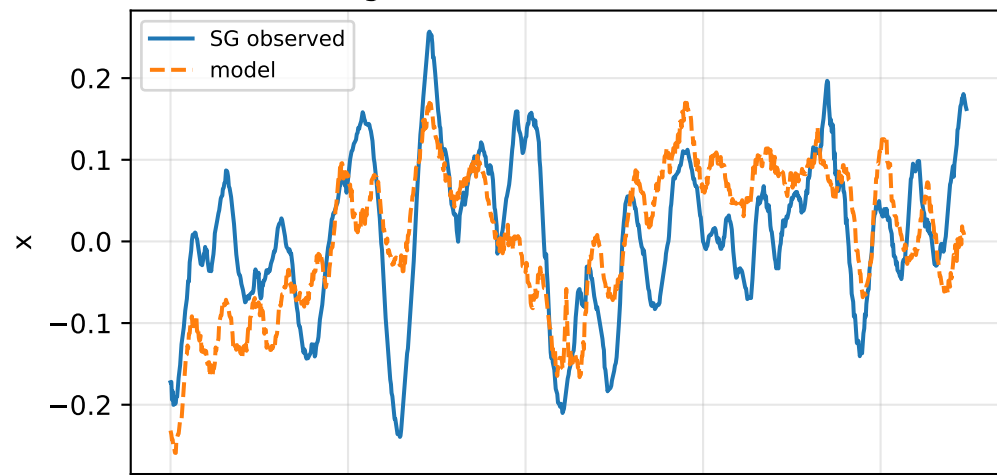


lag_legacy_smooth_schedule: acceleration residual objective

specific acceleration [m/s^2]

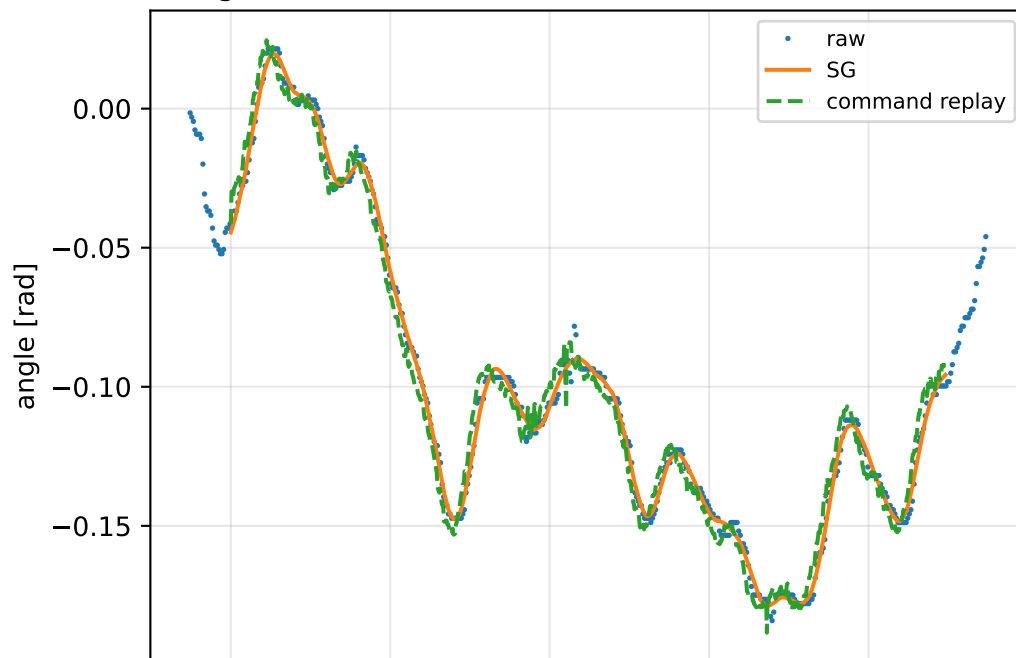


angular acceleration [rad/s^2]

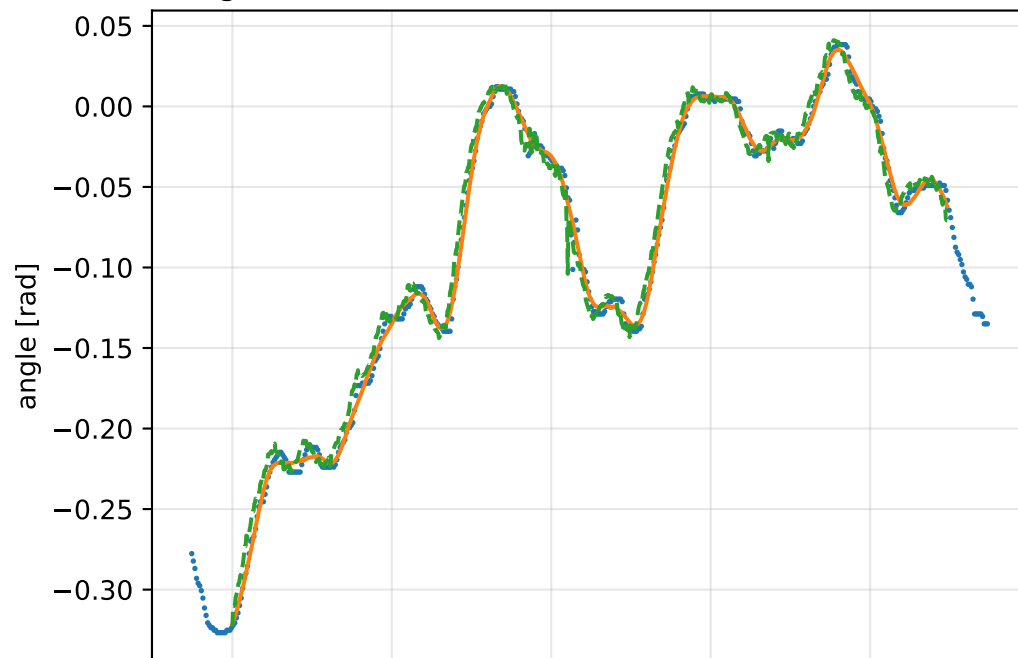


lag_legacy_smooth_schedule: gimbal measurement audit (objective=measured_sg)

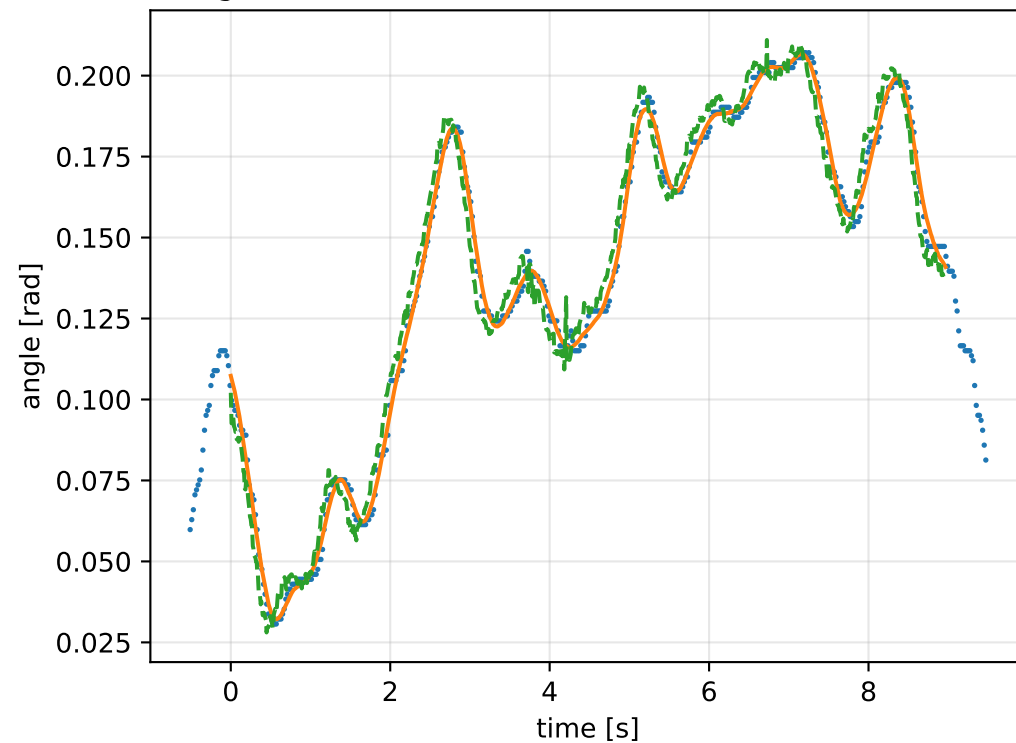
gimbal 1: RMSE=0.006345, max=0.01494 rad



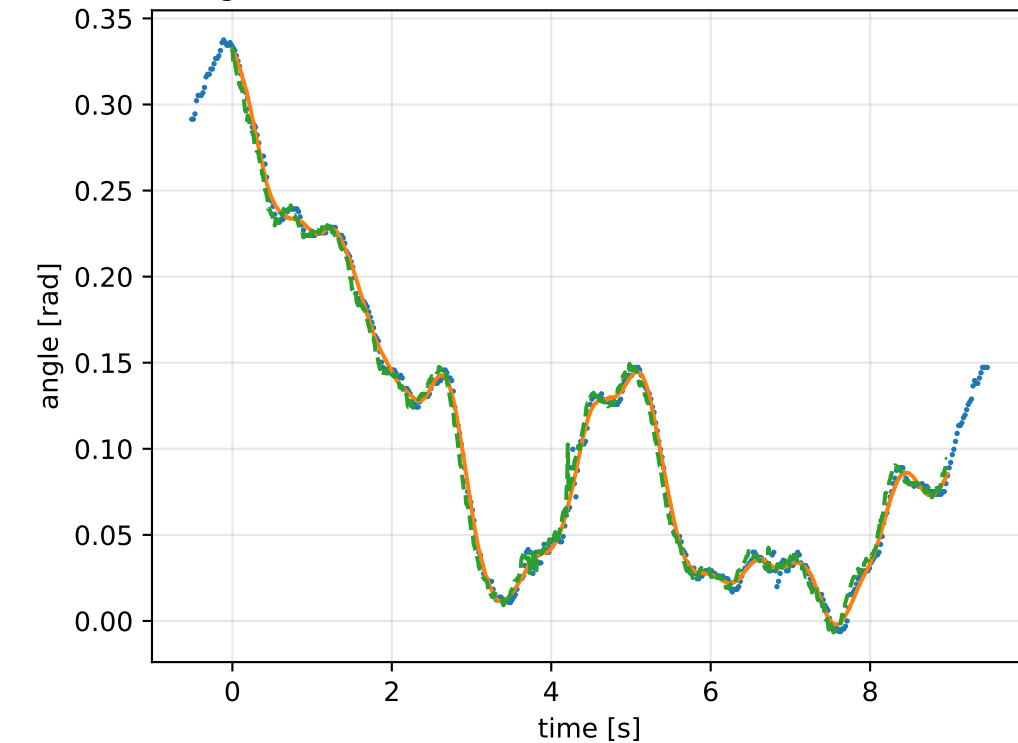
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

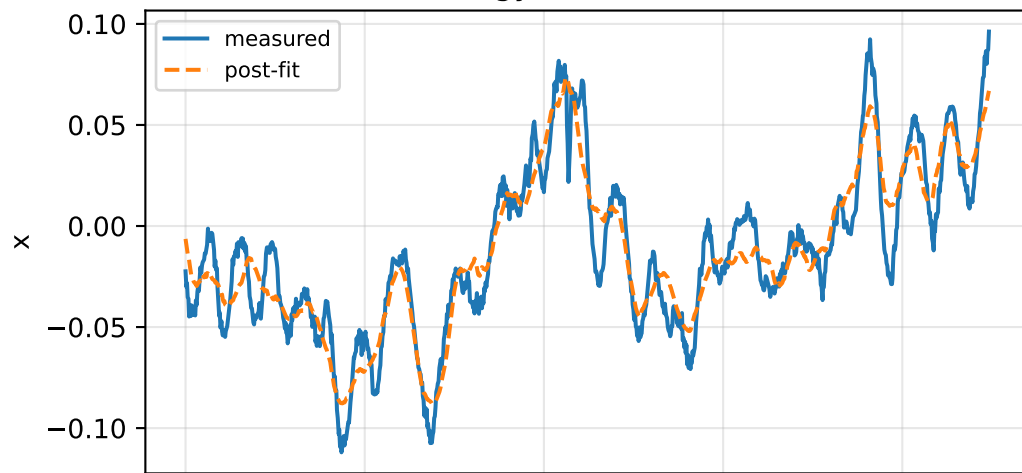


gimbal 4: RMSE=0.006321, max=0.03642 rad

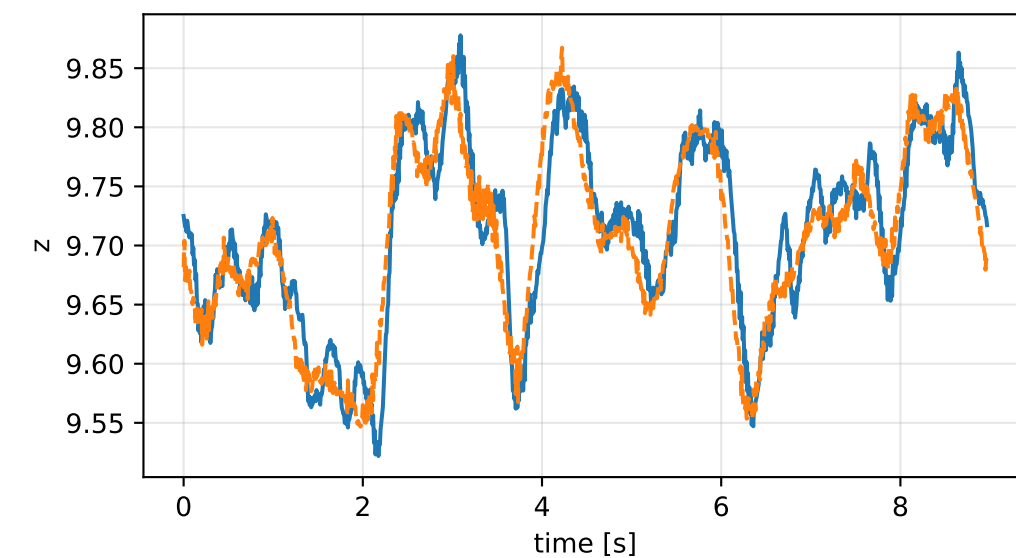
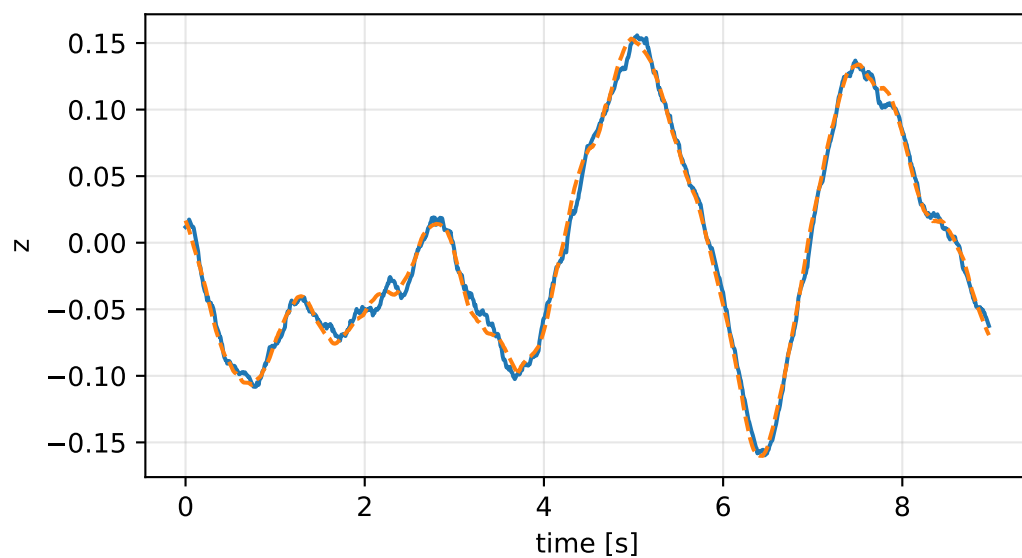
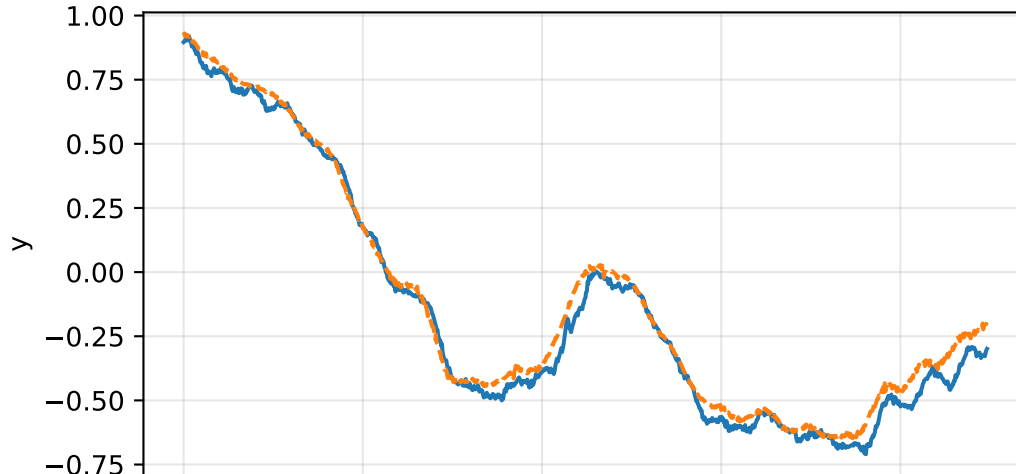
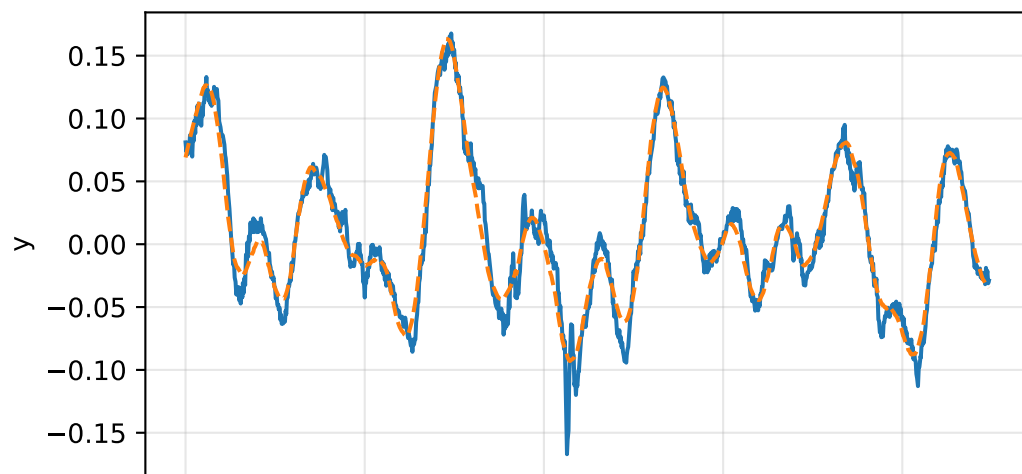
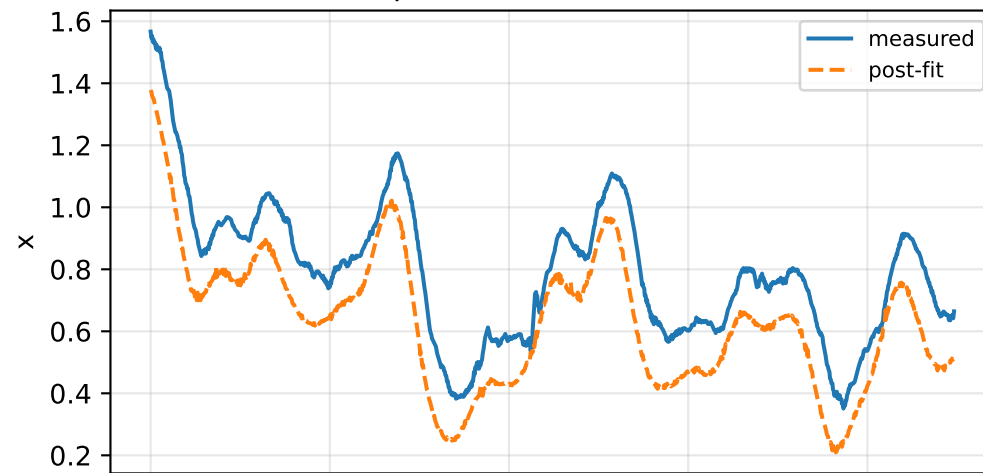


lag_legacy_smooth_schedule: IMU comparison (diagnostic only)

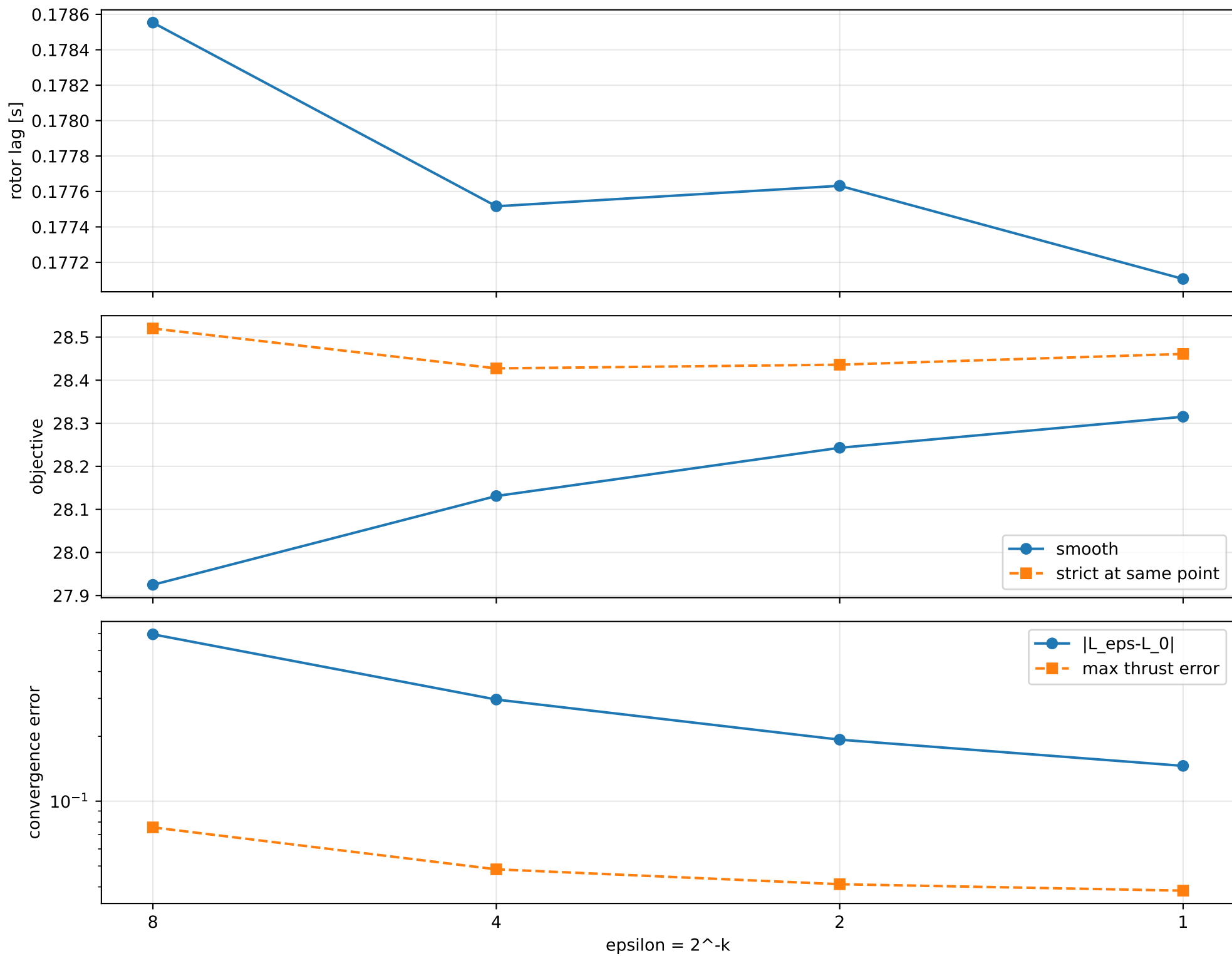
gyro [rad/s]



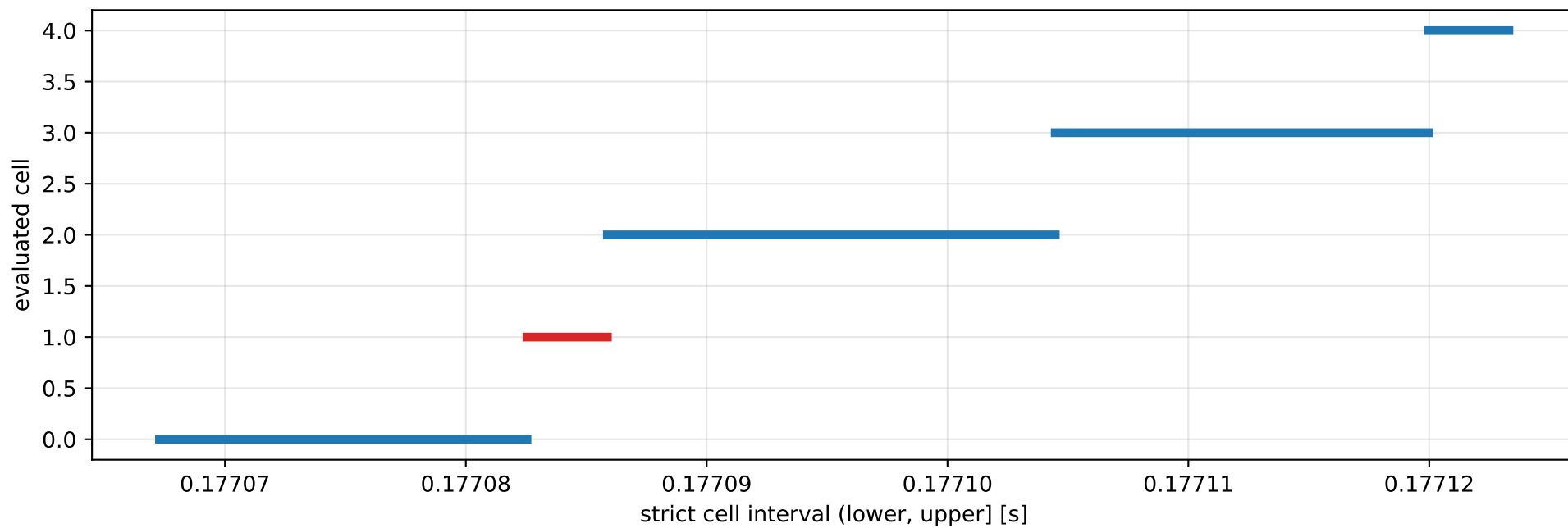
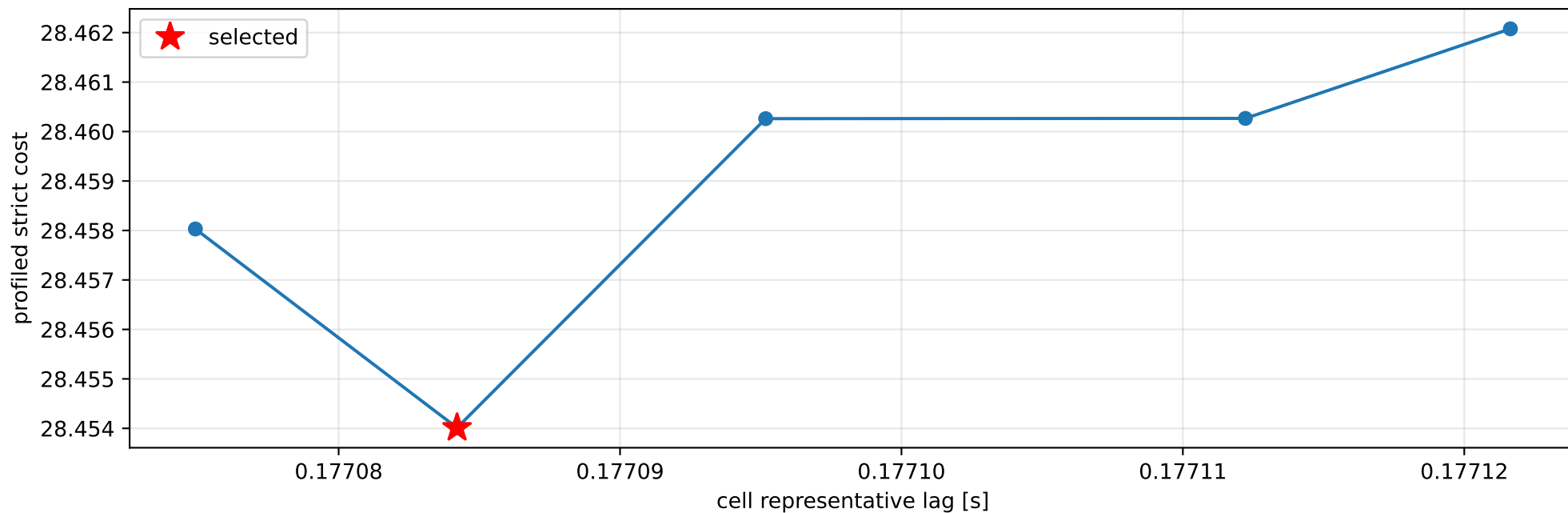
specific force [m/s^2]



lag_legacy_smooth_schedule: smooth-to-strict continuation



lag_legacy_smooth_schedule: exact strict-ZOH lag cells



selected (0.177082539, 0.177085876] s; final smooth 0.177106151 s; data boundary=False

lag_legacy_smooth_schedule: parameter estimate

Case: lag_legacy_smooth_schedule

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.65433841

inertia [kg m²]:

```
[[ 0.84876318 -0.01766289  0.06521403]
 [-0.01766289  0.4010378  -0.0115135 ]
 [ 0.06521403 -0.0115135  0.47125415]]
```

CoG body [m]: [-0.01975597 -0.05511542 -0.00494988]

force effectiveness: [1.62421713 1.58973412 0.83927929 1.01802079]

scale-free J/m [m²]:

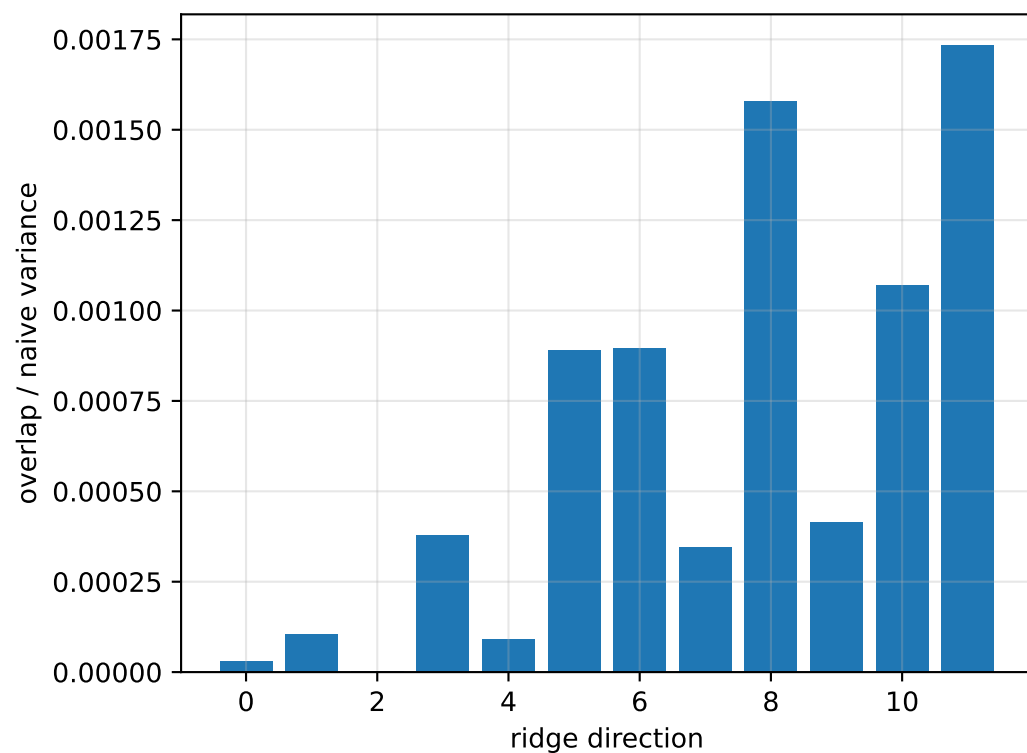
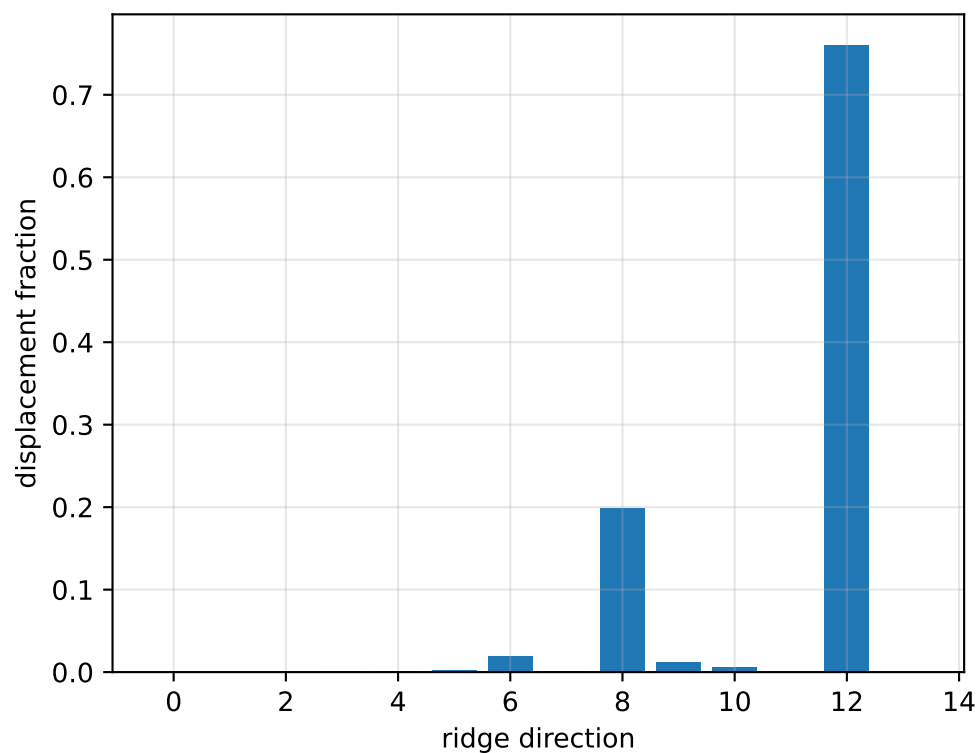
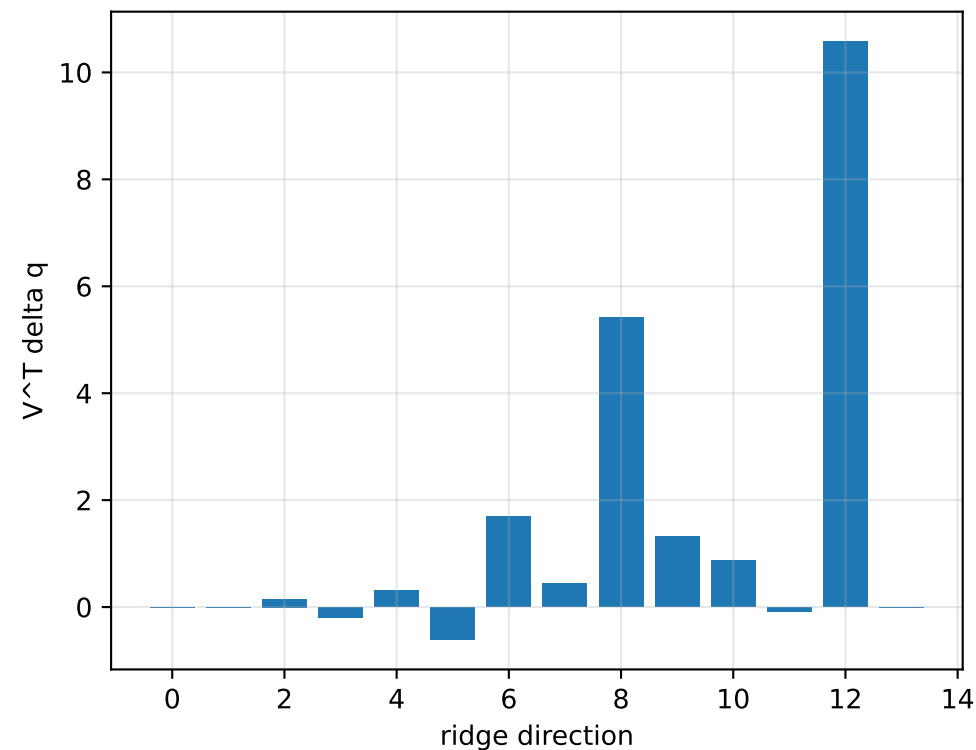
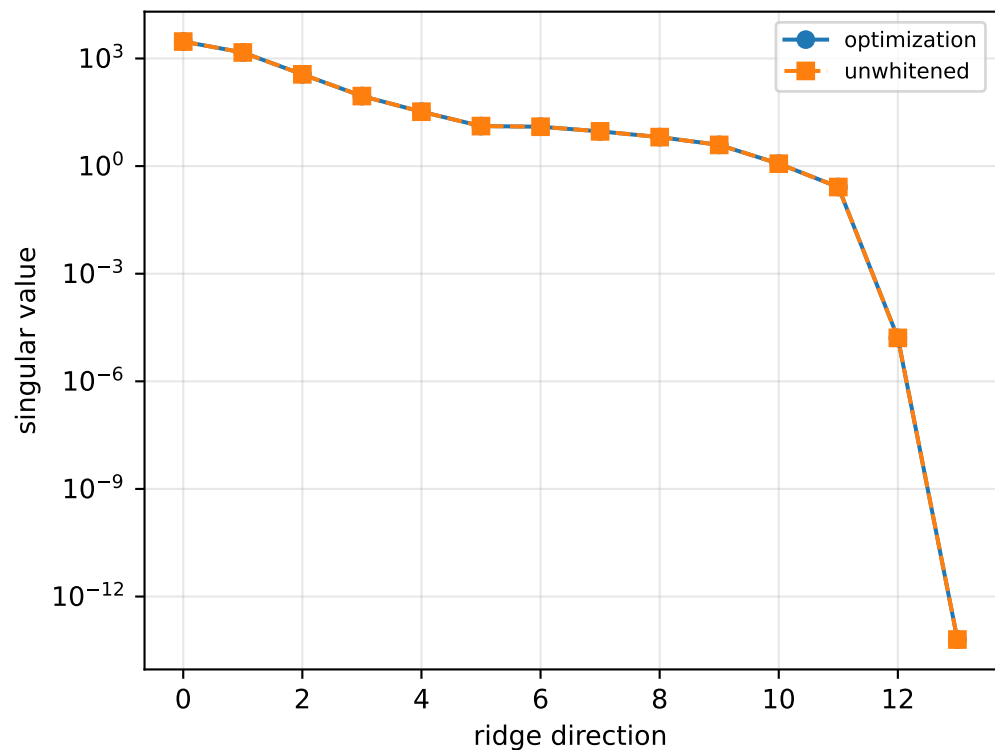
```
[[ 0.23226179 -0.0048334  0.01784565]
 [-0.0048334  0.10974293 -0.00315064]
 [ 0.01784565 -0.00315064  0.12895745]]
```

scale-free f/m: [0.4444627 0.43502652 0.22966655 0.27857868]

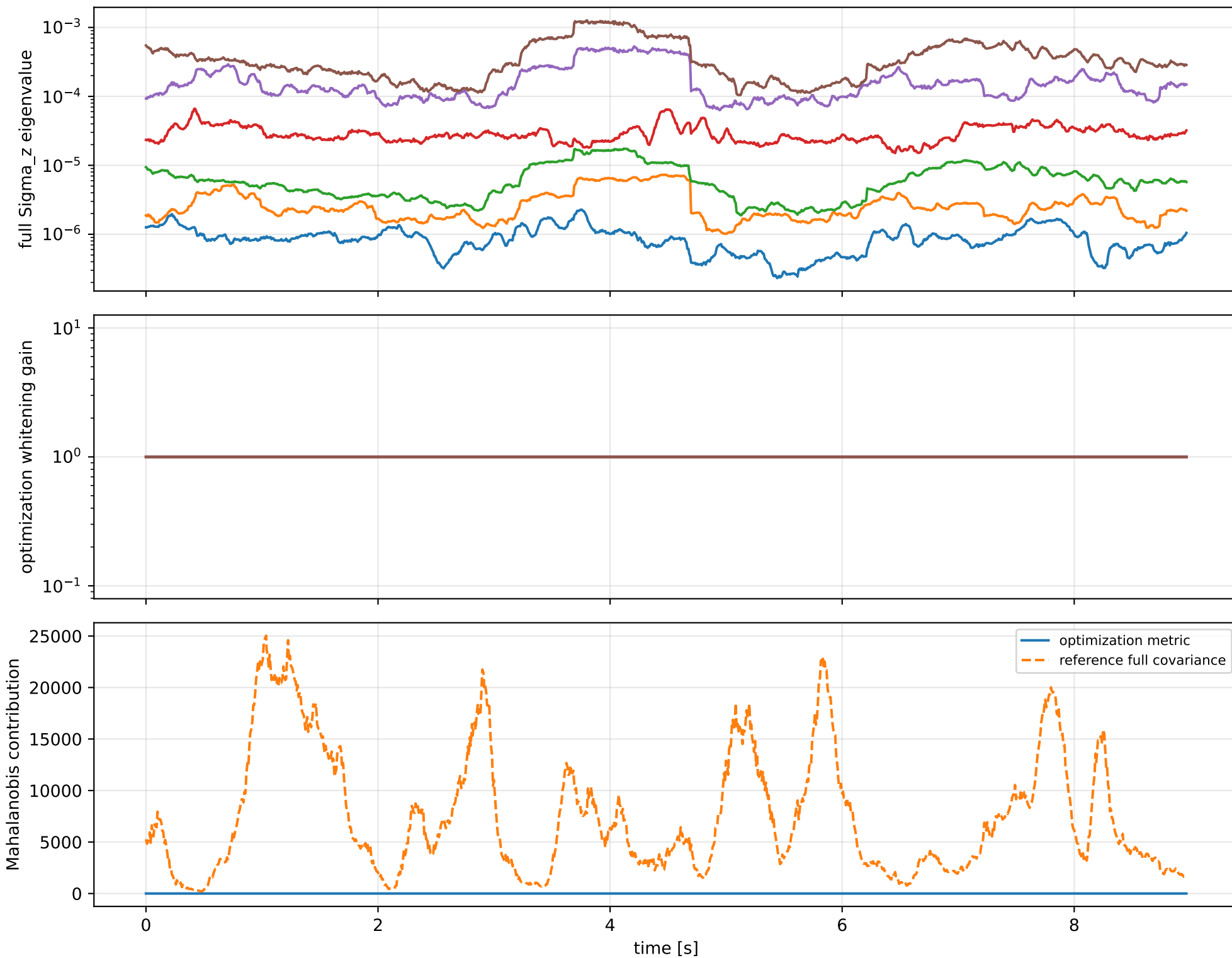
rotor lag cell: (0.177082539, 0.177085876] s

representative: 0.177084208 s

lag_legacy_smooth_schedule: ridge spectrum (rank 13, nullity 1, ||Jv||=7.95e-14)



lag_legacy_smooth_schedule: optimization vs reference covariance



lag_legacy_smooth_schedule: wrench / closure (force max 1.11e-14, torque max 6.94e-17)

